

Predicting Individual Stock Price Movements in the KOSPI Market

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1. Overview

This study aims to predict the direction of stock price movements in the KOSPI market by combining 20 years of historical price and trading volume data with sentiment indicators.

2. Data Collection and Preprocessing

- **Data:**
 - Daily OHLCV (Open, High, Low, Close, Volume) data for stocks in the KOSPI market over the past 20 years
 - Major economic news articles and earnings announcements related to each stock during the same period
- **Technical Indicators:**
Financial domain knowledge reflected through indicators such as Simple Moving Average (SMA), Relative Strength Index (RSI), MACD, and Bollinger Bands
- **Return Transformation:**
Log returns are used to improve data stationarity
- **Normalization:**
Min-Max Scaling is applied to reduce price-level differences across stocks

3. Research Methodology

1. Convert financial news into sentiment scores using FinBERT
2. Combine stock price data and sentiment scores into time-series input vectors
3. Train and compare ARIMA, LSTM, and Transformer-based models
4. Improve model performance through hyperparameter tuning

4. Performance Evaluation Metrics and Expected Outcomes

- **Evaluation Metrics:**
 - Classification Metrics: Accuracy, F1-score
 - Financial Metrics: Cumulative return and Sharpe ratio of a simulated portfolio based on prediction results
- **Expected Outcomes:**
 - Analyze how technical indicators affect stock price direction in the KOSPI market
 - Identify differences in prediction performance across stocks in the KOSPI market
 - Examine whether abrupt changes in sentiment serve as leading signals of higher volatility or trend reversals
 - Reflect event risks, such as earnings surprises or negative news, that are difficult to capture using stock price data alone